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Questions Based on Assignments :

Assignment B-1 : Predict the price of the Uber ride

1. **What is data preprocessing?**
   Data preprocessing is the process of cleaning and transforming raw data to prepare it for analysis, ensuring it's complete and ready for machine learning.

2. **Define Outliers.**
   Outliers are data points that are significantly different from most other data, which can affect analysis accuracy.

3. **What is Linear Regression?**
   Linear Regression is a statistical method that models the relationship between two variables by fitting a straight line to the data, predicting the dependent variable based on the independent variable.

4. **What is Random Forest Algorithm?**
   Random Forest is an ensemble learning algorithm that creates multiple decision trees and combines their results to improve prediction accuracy and reduce overfitting.

5. **Explain: pandas, numpy.**

- **Pandas:** A Python library for data manipulation and analysis, providing tools to work with data in tables.
- **NumPy:** A library for numerical computing in Python, used for working with arrays and performing mathematical operations.

Assignment B-2 : Classify the email using the binary classification method

1. **Data Preprocessing**
   Data preprocessing involves cleaning, transforming, and organizing raw data to make it suitable for analysis or machine learning models.

2. **Binary Classification**
   Binary classification is a type of classification where there are only two possible outcomes, such as "yes" or "no," "true" or "false."

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3. **K-Nearest Neighbours (K-NN)**
   K-NN is a simple algorithm that classifies a data point based on the majority class among its closest K neighbors in the dataset.

4. **Support Vector Machine (SVM)**
   SVM is a supervised learning algorithm that finds the best boundary (hyperplane) to separate data into different classes with maximum margin.

5. **Train, Test, and Split Procedure**
   This process divides a dataset into training and testing sets, where the training set trains the model, and the test set evaluates its performance.

Assignment B-3 : :Given a bank customer, build a neural network-based classifier that can determine whether they will leave or not in the next 6 months

1. **Artificial Neural Network (ANN)**
   ANN is a computational model inspired by the human brain, consisting of interconnected nodes (neurons) that learn patterns in data for tasks like classification and prediction.

2. **Keras**
   Keras is a high-level neural network library in Python, built on top of TensorFlow, that simplifies building and training deep learning models.

3. **TensorFlow**
   TensorFlow is an open-source machine learning framework that supports building, training, and deploying large-scale machine learning and deep learning models.

4. **Normalization**
   Normalization is the process of scaling data to a standard range (usually 0 to 1) to ensure each feature contributes equally to model performance.

5. **Confusion Matrix**
   A confusion matrix is a table that shows the performance of a classification model by comparing actual vs. predicted values for each class, helping to evaluate accuracy, precision, and recall.

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1. **Accuracy**

   Accuracy is the percentage of correct predictions out of all predictions made. It's calculated as:

$$\text{Accuracy} = \frac{\text{True Positives} + \text{True Negatives}}{\text{Total Predictions}}$$

2. **Precision**

   Precision is the percentage of true positive predictions out of all positive predictions made. It shows how accurate positive predictions are:

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

3. **Recall**

   Recall is the percentage of true positive predictions out of all actual positives. It shows how well the model captures actual positive cases:

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

Assignment B-4 : Implement K-Nearest Neighbors algorithm on diabetes.csv+

1. **What is data preprocessing, and why is it important?**
   Data preprocessing prepares raw data by cleaning, transforming, and organizing it to improve model performance and accuracy.

2. **How do you identify outliers in a dataset?**
   Outliers can be identified using methods like the Z-score, IQR (Interquartile Range), or visualizations like box plots to detect unusual data points.

3. **What is correlation, and why is it checked?**
   Correlation measures the relationship between two variables. Checking it helps identify dependencies or multicollinearity, which can impact model effectiveness.

4. **Explain the K-Nearest Neighbors (KNN) algorithm.**
   KNN classifies data points based on the class of the K closest neighbors, making predictions based on majority voting among neighbors.

5. **What is the Random Forest algorithm?**
   Random Forest is an ensemble algorithm that creates multiple decision trees and aggregates their results to enhance prediction accuracy and reduce overfitting.

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6. **What is a confusion matrix, and how is it used?**
   A confusion matrix is a table showing actual vs. predicted values in classification. It helps measure performance metrics like accuracy, precision, recall, and F1-score.

7. **Define accuracy_score.**
   Accuracy score is the percentage of correct predictions out of all predictions.

8. **What is mean_squared_error, and why is it used?**
   Mean Squared Error (MSE) measures the average squared difference between predicted and actual values, evaluating model accuracy in regression tasks.

9. **Explain r2_score and its significance.**
   R-squared (r2_score) indicates the proportion of variance explained by the model. It shows how well the model fits the data, with values closer to 1 being better.

10. **What is roc_auc_score, and why is it important?**
    The ROC AUC score measures a model's ability to distinguish between classes. It is the area under the ROC curve, where a score closer to 1 indicates better classification.

11. **Describe the ROC curve and its purpose.**
    The ROC curve is a plot of True Positive Rate vs. False Positive Rate for different thresholds, illustrating the trade-off between sensitivity and specificity in classification.

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Questions Based on Assignments :

Assignment C-1 : Installation of MetaMask and study spending Ether per transaction

1. **What is Blockchain**
   Blockchain is a decentralized, digital ledger technology that records transactions across multiple computers, ensuring transparency and security without a central authority.

2. **What is Cryptocurrency?**
   Cryptocurrency is a digital or virtual currency that uses cryptography for security, operates on blockchain technology, and enables peer-to-peer transactions without intermediaries, such as Bitcoin or Ethereum.

3. **What are Transaction Wallets?**
   Transaction wallets are digital wallets that store cryptocurrencies and allow users to send, receive, and manage their digital assets securely. Examples include MetaMask and Trust Wallet.

4. **What is an Ether Transaction?**
   An Ether transaction involves transferring Ether (ETH), the native currency of the Ethereum blockchain, between wallets. Transactions are recorded on the blockchain and require a small transaction fee.

5. **Installation Process of MetaMask**

- Step 1: Go to the MetaMask website or browser extension store.
- Step 2: Download and install MetaMask for your browser or mobile device.
- Step 3: Create a new wallet, set up a strong password, and securely store your backup phrase.
- Step 4: Once set up, MetaMask is ready to manage Ether and other Ethereum-based assets.

6. **What Are the Different Types of Blockchain Technology?**
   - **Public Blockchain:** Open to everyone (e.g., Bitcoin, Ethereum).
   - **Private Blockchain:** Restricted access for specific users.
   - **Consortium Blockchain:** Controlled by a group of organizations.

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- **Hybrid Blockchain:** Combines public and private features for flexibility.

7. **What Are the Key Features/Properties of Blockchain?**
   - **Decentralization:** No central authority; data is spread across nodes.
   - **Transparency:** Transactions are visible to all participants.
   - **Immutability:** Data cannot be altered once added to the blockchain.
   - **Security:** Cryptographic techniques secure data and transactions.

8. **What is the Difference Between Ethereum and Bitcoin?**
   - **Bitcoin:** Primarily a digital currency for secure payments.
   - **Ethereum:** A platform for creating decentralized applications (dApps) with smart contract functionality, in addition to its currency, Ether.

9. **What are Merkle Trees? Explain Their Concept.**
   Merkle Trees are a data structure used in blockchain to efficiently and securely verify transactions. They organize transaction data into a tree structure, where each leaf node is hashed, and hashes are combined up to form a single root hash, providing an easy way to verify data integrity.

10. **Give Real-Life Use Cases of Blockchain.**
    - **Finance:** Cross-border payments and decentralized finance (DeFi).
    - **Supply Chain:** Tracking goods from production to delivery.
    - **Healthcare:** Secure storage and sharing of patient records.
    - **Voting Systems:** Transparent and tamper-proof voting records.
    - **Digital Identity:** Managing digital identities and preventing fraud.

Assignment C-2 : Create your own wallet using Metamask.

1. **What is Bitcoin?**
   Bitcoin is a decentralized digital currency created in 2009 by an anonymous person or group known as Satoshi Nakamoto. It operates on blockchain technology, allowing peer-to-peer transactions without the need for intermediaries like banks.

2. **What Are the Biggest Four Common Cryptocurrency Scams?**

Phishing Scams: Fake websites or emails trick users into revealing private keys.
Ponzi Schemes: Fraudulent investment schemes promising high returns.
Fake ICOs (Initial Coin Offerings): Scammers create fake tokens or projects to get investors' funds.
Rug Pulls: Developers abandon a project suddenly, taking investors' money.

3. **Explain How Safe Are Money E-Transfers?**
   Money e-transfers can be safe when using secure networks and reputable services. Encryption and authentication protocols make them secure, but users should be cautious of phishing, hacking, or unauthorized access.

4. **What is Cryptojacking and How Does It Work?**
   Cryptojacking is unauthorized use of someone's computer or device to mine cryptocurrency. Attackers install malware or use malicious scripts on websites to hijack device resources, slowing down systems and potentially causing damage.

Assignment C-3 : : Write a smart contract on a test network, for Bank account

1. **What is a Smart Contract?**
   A smart contract is a self-executing contract with code that runs on a blockchain. It automatically enforces terms and conditions agreed upon by all parties**.**

2. **What is a Test Network?**
   A test network (testnet) is a blockchain network used for testing smart contracts and applications without real currency. Examples include Ropsten and Goerli for Ethereum.

3. **What is MetaMask, and why is it used?**
   MetaMask is a browser extension and wallet that allows users to manage and interact with Ethereum-based assets and decentralized applications (dApps). It's used to deploy and test smart contracts on test networks.

4. **Explain the Role of MetaMask in Smart Contract Deployment.**
   MetaMask connects to Ethereum networks and provides an account to sign and send transactions, allowing users to deploy and test smart contracts.

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5. **How Do You Deploy a Smart Contract on a Test Network?**
   - Write the smart contract code.
   - Use Remix (online IDE) to compile the code.
   - Connect MetaMask to Remix and choose a test network.
   - Deploy the contract using MetaMask to sign and send the transaction.

6. **Explain the Deposit Operation in the Bank Account Smart Contract.**
   The deposit function allows a user to send a specific amount of cryptocurrency to their account balance within the smart contract.

7. **Explain the Withdraw Operation in the Bank Account Smart Contract.**
   The withdraw function enables a user to take money out of their account balance, provided they have sufficient funds available.

8. **How Does the Show Balance Operation Work in the Smart Contract?**
   The show balance function returns the current balance of a user's account, showing how much cryptocurrency they have deposited in the contract**.**

9. **What Is the Objective of Using MetaMask and Test Network in This Assignment?**
   The objective is to teach students how to interact with Ethereum networks and deploy smart contracts, gaining practical experience in blockchain technology and its applications.

10. **What Are the Key Advantages of Using Smart Contracts for Banking Operations?**
    - Transparency: All operations are recorded on the blockchain.
    - Security: Transactions are encrypted and tamper-proof
    - Automation: Smart contracts self-execute based on coded conditions.

11. **How Do You Test and Verify the Smart Contract?**
    Use a test network to run transactions (deposit, withdraw, balance check) and confirm they work as expected. Errors or bugs can be fixed and retested

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12. **What Are Gas Fees, and Why Are They Relevant in Smart Contracts?**
    Gas fees are the transaction costs required to run a smart contract on the blockchain. They're necessary for processing and securing transactions but are lower on test networks.

Assignment C-4 : Solidity

1. **What is Solidity?**
   Solidity is a high-level programming language designed for writing smart contracts on the Ethereum blockchain, enabling developers to create decentralized applications.

2. **What is a Smart Contract?**
   A smart contract is a self-executing contract with terms directly written into code, running on a blockchain, which automatically enforces the agreed-upon conditions

3. **What is the Purpose of Creating Student Data in a Smart Contract?**
   The purpose is to store and manage student information securely on the blockchain, allowing for transparent and tamper-proof records**.**

4. **What are Structures in Solidity?**
   Structures (structs) are user-defined data types that allow developers to group related variables together, such as a student's name, age, and grade.

5. **What are Arrays in Solidity?**
   Arrays are data structures that can store multiple values of the same type, allowing for easy access and manipulation of collections of data, like a list of students.

6. **What is the Fallback Function in Solidity?**
   The fallback function is a default function that is called when a smart contract is sent Ether without data or when no other function matches the call, allowing the contract to handle unexpected transactions

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Questions Based on Assignments :

Assignment A-1 : Fibonacci series

1. **What is the Fibonacci Sequence of numbers?**
   The Fibonacci Sequence is a series where each number is the sum of the two preceding ones, starting from 0 and 1 (e.g., 0, 1, 1, 2, 3, 5, 8...).

2. **How do the Fibonacci numbers work?**
   Each Fibonacci number is derived by adding the previous two numbers, following the formula $F(n)=F(n-1)+F(n-2)$ with $F(0)=0F(0) = 0F(0)=0$ and $F(1)=1F(1) = 1F(1)=1$.

3. **What is the Golden Ratio?**
   The Golden Ratio, approximately 1.618, is the limit of the ratio of consecutive Fibonacci numbers, representing an ideal proportion in art, nature, and architecture.

4. **What is the Fibonacci Search technique?**
   Fibonacci Search is a search algorithm for sorted arrays, using Fibonacci numbers to split the array, which reduces comparisons and is more efficient for larger datasets.

5. **What is the real application for Fibonacci series?**
   The Fibonacci series is used in algorithm design, financial modeling, biological studies (like branching in trees and leaf arrangements), and computer data structures.

Assignment A-2 : Huffman Encoding

1. **What is Huffman Encoding?**

Huffman Encoding is a compression technique that assigns shorter binary codes to more frequent characters and longer codes to less frequent ones, reducing the overall size of data.

2. **How many bits may be required for encoding the message ‘mississippi’?**

Encoding 'mississippi' requires 28 bits, with shorter codes for frequently occurring letters like 'i' and 's'.

3. **Which tree is used in Huffman encoding? Give one Example**

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A binary tree is used in Huffman encoding. For example, in encoding 'mississippi', 'i' and 's' (most frequent) have the shortest codes, while 'm' and 'p' (least frequent) have longer codes.

4. **Why is Huffman coding lossless compression?**

Huffman coding is lossless because it preserves the exact original data, allowing it to be perfectly reconstructed from the compressed data.

Assignment A-3 : fractional Knapsack problem using Greedy

1. **What is Greedy Approach?**
   The Greedy Approach is an algorithmic paradigm that builds a solution incrementally by choosing the locally optimal choice at each step, with the hope of finding a global optimum.

2. **Explain the concept of fractional knapsack.**
   The fractional knapsack problem allows the breaking of items into smaller pieces. Given a set of items, each with a weight and value, the goal is to maximize the total value in the knapsack, where items can be divided.

3. **Difference between Fractional and 0/1 Knapsack.**
   In the Fractional Knapsack, items can be divided into smaller parts, while in the 0/1 Knapsack, items must be taken whole or not at all. This results in different approaches and solutions for each problem.

4. **Solve one example based on Fractional Knapsack.**

   Items:
   - Item 1: Weight = 10 kg, Value = 60
   - Item 2: Weight = 20 kg, Value = 100
   - Item 3: Weight = 30 kg, Value = 120

   Knapsack Capacity: 50 kg

➔ **Solution:**

1. **Calculate value-to-weight ratio:**
   - Item 1: 6 per kg
   - Item 2: 5 per kg
   - Item 3: 4 per kg
2. **Sort items by ratio: Item 1, Item 2, Item 3**.
3. **Fill the knapsack:**
   - Take Item 1 (10 kg, 60).
   - Take Item 2 (20 kg, 100).
   - Take half of Item 3 (15 kg, 60).
4. **Total value = 60 + 100 + 60 = 220.**

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Assignment A-4 : 0-1 Knapsack problem using dynamic programming

1. **What is Dynamic Approach?**
   An algorithmic technique that breaks a problem into simpler subproblems, storing results to avoid redundant calculations.
2. **Explain the concept of 0/1 knapsack.**
   A problem where you select items with given weights and values to maximize total value without exceeding a weight limit, with each item being either included or excluded.
3. **Difference between Dynamic and Branch and Bound Approach. Which is best?**
   Dynamic Programming stores results of overlapping subproblems, while Branch and Bound explores branches and prunes unpromising ones. The best approach depends on the problem context.
4. **Solve one example based on 0/1 knapsack.**
   → **Example:**
   - **Items:**
   - Item 1: Weight = 1 kg, Value = 1
   - Item 2: Weight = 3 kg, Value = 4
   - Item 3: Weight = 4 kg, Value = 5
   - Item 4: Weight = 5 kg, Value = 7
   - **Knapsack Capacity: 7 kg**

   Solution:
   Maximum value is 8, achieved by including Item 4 (Weight = 5 kg, Value = 7) and Item 1 (Weight = 1 kg, Value = 1).

Assignment A-5 : n-Queens matrix

1. **What is backtracking? Give the general Procedure.**
   Backtracking is a method for solving problems by trying different options and going back if a choice doesn't work. The general procedure is to explore all possible solutions, check if they are valid, and undo choices when needed.
2. **Give the problem statement of the n-queens problem. Explain the solution.**
   The n-queens problem asks how to place n queens on an n × n chessboard so that no two queens can attack each other. The solution involves placing queens one row at a time and making sure that no two queens are in the same row, column, or diagonal.

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3. **Write an algorithm for N-queens problem using backtracking.**

- Initialize an empty N x N chessboard.
- Check if placing a queen at (row, col) is safe.
- If all queens are placed, print the board.
- For each row in the current column:
- If safe, place a queen and move to the next column.
- If not successful, remove the queen (backtrack).
- Start with the first column.

4. **Why is it applicable to N=4 and N=8 only?**

While backtracking can be used for any N, N=4 and N=8 are popular examples because they are easier to understand and visualize. They help demonstrate how the backtracking method works effectively**.**